# DERIVATIVES AND SHAPE QUIZ

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Name: Solutions

**1.** For the function

$$f(x) = \frac{x^2}{x-4}$$

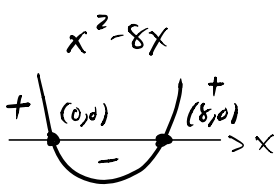
find

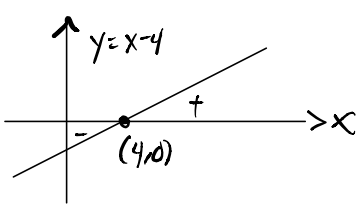
- the interval(s) where the function is increasing/decreasing,
- the interval(s) where the function is concave up/down
- the local maximum and minimum values,
- the inflection points.

$$f'(x) = \frac{2x(x-4) - x^2}{(x-4)^2} = \frac{2x^2 - 8x - x^2}{(x-4)^2} = \frac{x^2-8x}{(x-4)^2}$$

$$f''(x) = \frac{(2x-8)(x-4)^2 - (x^2-8x)(2)(x-4)(1)}{(x-4)^4} = \frac{2(x-4)^3 - 2(x^2-8x)(x-4)}{(x-4)^4}$$

$$= \frac{2(x-4)^2 - 2(x^2-8x)}{(x-4)^3} = \frac{2x^2-16x+32-2x^2+16x}{(x-4)^3} = \frac{32}{(x-4)^3}$$




$(x-4)^2 \geq 0 \Rightarrow$ sign of $x^2-8x$ determines sign of $f'(x)$

and $x-4$ determines sign of $(x-4)^3$, hence also the sign of $f''(x)$:

Increasing: $(-\infty,0) \cup (8,\infty)$ Concave up: $(4,\infty)$

Decreasing: $(0,4) \cup (4,8)$ Concave down: $(-\infty,4)$

local max: $(0,0)$ inflection point: None

local min: $(8,16)$ ($x=4$ is not in domain of $f$).

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